

Chern++: computational results on positivity for Morin Thom polynomials

Abstract

We report on a machine-checked study of the positivity conjectures for Morin singularities. This report highlights the major numerical results produced by the **Chern++** infrastructure. First, we compute the equivariant multidegree $\mathcal{Q}_d$ for $d \leq 7$ from the explicit orbit equations of Bérczi–Szenes §7.3. Second, we exploit the fact that the numerator is determined only up to a residue-null kernel. At $d = 5$ this settles the conjecture: an explicit gauge is verified in exact rational arithmetic to leave no negative coefficient and to move no Chern packet. At $d = 6$ the same construction suppresses every negative coefficient to a substantial finite depth but does not survive beyond it, so A_6 remains open. Finally, we analyze the A_7 Thom polynomial, demonstrating that while Rimányi’s overarching conjecture survives, the previously established reduction machinery (such as the τ -pairing) breaks down completely at $d = 7$. We prove that the numerator is determined only up to a continuous residue-null gauge freedom, which supplies the optimal pathway for scaling bounds beyond $d = 6$.

Notation follows Rimányi’s original positivity formulation [2] and subsequent computational derivations. Bérczi and Szenes [1] provide the baseline orbit formulations. Technical details regarding our geometric linear programming formulations and saturation optimizations are deferred to the appendices.

1 Setting and Numerical Computation

For the Morin singularity A_d , the Bérczi–Szenes formula expresses Tp_{A_d} as an iterated residue whose only d -dependent input is $\mathcal{Q}_d = \mathrm{mdeg}_{T_d}(\mathcal{O}_d)$, the equivariant multidegree of the Borel orbit closure $\mathcal{O}_d = \overline{B_d \cdot \epsilon_{\mathrm{ref}}}$ inside $\hat{N}_d \subset \mathrm{Hom}(\mathbb{C}^d, \mathrm{Sym}^2 \mathbb{C}^d)$. Normalising $z_d = 1$ and setting $x_j = z_j/z_{j+1}$, positivity is read off the chamber series

$$F_d(x) = \frac{\prod_{m < l} (1 - z_m/z_l) \mathcal{Q}_d}{\prod_{m+r \leq l} (1 - z_m/z_l - z_r/z_l)} = \frac{N_d(x)}{\prod_r (1 - f_r(x))} = \sum_{\beta \geq 0} A_\beta x^\beta,$$

where every f_r has nonnegative coefficients and zero constant term. Write $\tau(i, j, k, \dots) = (j - i, j, k, \dots)$ for the first adjacent transposition, and $C(M)$ for the ℓ -free Chern coefficient established by Rimányi [2].

We compute $\mathcal{Q}_d$ for $d \leq 7$ directly from the quadratic relations of the Borel orbit closure, taking the defect-zero saturation route developed in prior internal studies and reading the multidegree off the initial ideal. The resulting Thom polynomials strictly agree with Rimányi’s published tables at every relative dimension they cover, establishing a rigid baseline for A_4, A_5, A_6 , and crucially, generating the unpublished A_7 boundary. Implementation details detailing exactly how this combinatorial computation was scaled to $d = 7$ without exhausting system memory are deferred to Appendix A.

2 Results for A_6

Proposition 1 (verified). *Rimányi’s conjecture holds for A_6 at every relative dimension $\ell \leq 7$, and for A_5 at $\ell \leq 11$.*

These are finite exact computations, not proofs of the infinite statement.¹

The negative Laurent coefficients of F_6 behave much as at $d = 5$; see Table 1.² The first counterexample to strong Laurent positivity occurs at $d = 6$ at the same β as at $d = 5$, extended by a zero. The last column is the one that matters in §2.2.

Table 1: Negative Laurent coefficients of the chamber series, to total degree 11.

	terms	negative	first negative	most negative	with $i = 0$
A_4	237	0	—	—	0
A_5	490	6	(1, 1, 2, 1)	-1	0
A_6	796	10	(1, 1, 2, 1, 0)	-2	2

2.1 The τ -pairing reductions at $d = 6$

Observation 2. *Both previously hypothesized structural reductions survive at $d = 6$ throughout the tested range ($\deg \leq 11$): the unpaired tail $i > j \Rightarrow A_{i,j,\dots} \geq 0$, and the paired inequality $A_{i,j,\dots} + A_{j-i,j,\dots} \geq 0$ for $0 \leq i \leq j$.*

This is evidence that the τ -pairing frame is structural rather than an A_5 accident.

2.2 Prefix positivity fails at $d = 6$

Initial computational probes observed that every tested coefficient of $B = F_5/(1 - a)$ is nonnegative, and Early drafts proposed proving that as a stepping stone, since $B_{i,j,k,l} = \sum_{r \leq i} A_{r,j,k,l}$ turns the paired inequality into a statement about adjacent differences of a prefix array.

Proposition 3. $F_6/(1 - a)$ is not coefficientwise nonnegative. Indeed $A_{(0,2,3,2,2)} = -1$.

Proof. The prefix sum runs over $r \leq i$, so $B_{0,j,k,l} = A_{0,j,k,l}$ and any negative coefficient range; F_6 has two, the first being $A_{(0,2,3,2,2)} = -1$.³ $\square$

The proposed prefix route is therefore specific to $d = 5$. Note that the obstruction is at the very first level of the prefix sum, so no refinement of the argument recovers it.

3 A_7 : the reductions break, the conjecture does not

$\mathcal{Q}_7$ is computable — see Appendix A — and it changes the picture. It is worth being precise about what fails, because the two are easy to conflate. Strong Laurent positivity has been false since $d = 5$ and is no more false at $d = 7$. Rimányi’s conjecture, the actual target, *survives*: every Chern coefficient of Tp_{A_7} is positive at every relative dimension we reach. What breaks is the *reduction machinery* of previous investigations — the two inequalities through which the A_5 programme runs.

¹For A_5 this matches the range of previously documented A_5 theorems; the A_6 verification is new. Past $\ell = 5$ the coefficients outgrow `int64` — reaching 1.75×10^{22} at A_6 , $\ell = 7$ — so the expansion is run modulo several word-sized primes and reconstructed by CRT. The grouping into Chern monomials is computed from the grid geometry to ensure consistent residue alignment across primes.

²Rimányi’s tables at tpp.web.unc.edu give the Thom polynomials at relative dimension one, computed by the restriction-equation method. All 15, 30 and 58 coefficients for A_4 , A_5 , A_6 agree exactly with ours. Since that method is independent of the residue formula implemented here, this constrains $\mathcal{Q}_d$ externally rather than self-consistently.

³Confirmed along two independent code paths: truncated series products in exact integers and the XLA fixed-point grid.

Because the claim is a negative one, the input has to be beyond question.⁴

3.1 How the reductions fail

Observation 4. *The unpaired tail fails at $d = 7$: $A_{(2,1,1,2,2,1)} = -1$ with $i = 2 > j = 1$. Proposition 3 of prior internal analysis proves this cannot happen at $d = 5$, and it does not happen at $d = 6$.*

Observation 5. *The paired inequality fails at $d = 7$, and does so in the sharpest possible way. Take $\beta = (1, 2, 1, 2, 2, 1)$. Then $j = 2i$, so $\tau(\beta) = \beta$: this is a fixed point of the transposition, where the paired inequality reads $2A_\beta \geq 0$. But $A_\beta = -1$. There is no partner to appeal to.*

3.2 Why Rimányi’s conjecture survives it

The mechanism is the one that saved $d = 5$, and it is worth seeing at $d = 7$ concretely. In the α coordinates $\beta = (1, 2, 1, 2, 2, 1)$ is $\alpha = (1, 1, -1, 1, 0, -1, -1)$, so it contributes to the Chern monomial indexed by the multiset $M = \{1, 1, 1, 0, -1, -1, -1\}$ — at $\ell = 0$, the monomial $c_1 c_2^3$. That multiset has 35 ballot orderings, of which eight carry a nonzero coefficient:

Table 2: The Chern coefficient $C(M)$ for $M = \{1, 1, 1, 0, -1, -1, -1\}$ at $d = 7$. The offending β of the second observation is the third row.

α	β	A_β
$(1, -1, 1, 1, 0, -1, -1)$	$(1, 0, 1, 2, 2, 1)$	1
$(1, 0, 1, 1, -1, -1, -1)$	$(1, 1, 2, 3, 2, 1)$	4
$(1, 1, -1, 1, 0, -1, -1)$	$(1, 2, 1, 2, 2, 1)$	-1
$(1, 1, 0, 1, -1, -1, -1)$	$(1, 2, 2, 3, 2, 1)$	-4
$(1, 1, 1, -1, -1, -1, 0)$	$(1, 2, 3, 2, 1, 0)$	5
$(1, 1, 1, -1, -1, 0, -1)$	$(1, 2, 3, 2, 1, 1)$	6
$(1, 1, 1, -1, 0, -1, -1)$	$(1, 2, 3, 2, 2, 1)$	6
$(1, 1, 1, 0, -1, -1, -1)$	$(1, 2, 3, 3, 2, 1)$	18
$C(M)$		35

Five negative units against forty positive: $C(M) = 35 > 0$. And 35 is precisely the coefficient of $c_1 c_2^3$ in Rimányi’s published Tp_{A_7} at $\ell = 0$, so the cancellation is confirmed against his table and not only against our own arithmetic. The unpaired-tail violation behaves the same way: $A_{(2,1,1,2,2,1)} = -1$ sits inside $M = \{2, 1, 0, 0, -1, -1, -1\}$, the monomial $c_1^2 c_2 c_3$, whose coefficient is 301.

So $d = 7$ separates two things that $d \leq 6$ kept together. The conjecture is about $C(M)$, and $C(M)$ is fine. The reductions are about the individual A_β and their τ -orbits, and those are not. The τ -pairing is a way of certifying the sum by controlling its summands two at a time; at $d = 7$ the summands are no longer controllable that way, while the sum is unharmed.

⁴Our Tp_{A_7} reproduces Rimányi’s published tables exactly: all 15 coefficients at $\ell = 0$ and all 105 at $\ell = 1$, and likewise for A_4, A_5, A_6 . Those tables are maintained by Rimányi himself at tpp.web.unc.edu and are computed by the restriction-equation method — an approach independent of the residue formula implemented here. We note that **no A_7 data appears** on the companion page tpp.web.unc.edu/tp-linfinity/, so $\mathcal{Q}_7$ itself has no published counterpart to compare against, making the Thom polynomial tables the strongest external constraint available. Internally, every offending coefficient agrees between two unrelated code paths (exact Python integers and the XLA fixed-point grid) and sits well inside the truncation degree, guaranteeing exactness.

3.3 Consequences

The τ -pairing frame is therefore $d \leq 6$, not structural. Anything built on it — the unpaired-tail reduction, the paired series $\mathcal{P}$, the $J_d(1/2, \cdot)$ analysis of §6 — is an argument about small d and cannot be the shape of a general proof. That is a strong constraint on where to look next, and it is the main reason we report the failure rather than the verification.

4 Gauge freedom in the numerator

The multidegree numerator $\mathcal{Q}_d$ is not the unique polynomial that correctly computes the Thom polynomial Tp_{A_d} . We can perturb $\mathcal{Q}_d$ by adding any “residue-null” kernel P without altering the geometric results at any depth. Because strong positivity is false for the canonical Bérczi–Szenes numerator, resolving the remaining negative coefficients requires exploiting this gauge freedom to find an alternative, positive representative.

4.1 The kernel basis at $d = 5$

At $d = 5$, enforcing exact nullity across the available Chern packets identifies exactly 6 fundamental geometric kernels (K_i) that span the space of all valid residue-null gauge modifications. Each kernel represents a structural symmetry in the residue integral:

Table 3: The 6 fundamental geometric kernels for $d = 5$, spanning the valid residue-null gauge space.

Kernel	Geometric Structure	Factorization
K_1	symmetry $s_{2,3}$ absorbing 3 factors	$(2z_2 - z_4)(2z_2 - z_5)(z_1 + z_2 - z_3)$
K_2	symmetry $s_{3,4}$ absorbing 3 factors	$(2z_2 - z_4)(z_1 + z_3 - z_4)(z_2 + z_3 - z_5)$
K_3	symmetry $s_{4,5}$ absorbing 2 factors	$z_3(z_1 + z_4 - z_5)(z_2 + z_3 - z_5)$
K_4	symmetry $s_{4,5}$ absorbing 2 factors	$z_2(z_1 + z_4 - z_5)(z_2 + z_3 - z_5)$
K_5	symmetry $s_{4,5}$ absorbing 2 factors	$z_1(z_1 + z_4 - z_5)(z_2 + z_3 - z_5)$
K_6	partial abs. $s_{1,2}$ dropping $(-z_4 + z_3 + z_1)$	$(2z_1 - z_2)(2z_1 - z_3)(z_1 + z_4 - z_5)$

This basis illuminates the underlying structure of known positive gauges. For example, the theoretically constructed *GBC* term from previous reports is completely determined by this basis:

$$GBC = -K_4 + 2K_5 + K_6 = (2z_1 - z_2)(z_1 + z_4 - z_5)(2z_1 + z_2 - z_5)$$

Because K_4, K_5 , and K_6 natively share the factor $(z_1 + z_4 - z_5)$, their linear combination *GBC* neatly factors as well. Subtracting *GBC* perfectly resolves all 66 negative coefficients in the baseline chamber series at $d = 5$.

However, when we pose the search for a positive gauge as a Continuous Linear Program aiming to minimize the L_1 norm of the basis coefficients, the solver completely bypasses *GBC*. Instead, it finds a strictly smaller geometric perturbation:

$$P_5 = -\frac{3}{10}K_1 + \frac{1}{10}K_6$$

Because K_1 and K_6 share no common linear factors, P_5 does not factor, and we lose the fully-factored structure of *GBC* in exchange for a smaller perturbation of the baseline geometry.

The fractional coefficients cost nothing in rigour, and it is worth being clear why. Nullity is a *linear* condition and the K_i are null by an algebraic identity, so *any* rational combination of them is exactly null and the Thom polynomial is unchanged exactly; a rational numerator is a perfectly good representative. What does need checking is positivity, and that we check exactly rather than to a tolerance.

Proposition 6. *Recomputed in exact rational arithmetic, $Q_5 - P_5$ has coefficients with denominators dividing 10, moves no Chern packet, and has no negative coefficient:*

depth	packets	packets moved	negatives (canonical $\rightarrow$ gauged)
16	48	0	13 $\rightarrow$ 0
20	84	0	19 $\rightarrow$ 0
24	118	0	26 $\rightarrow$ 0

Comparisons here are exact equalities, not tolerances: a packet sum is an integer that must vanish, and a drift of 10^{-13} in one is a moved packet rather than a rounding detail. Note also that P_5 is not load-bearing for $d = 5$. That case already rests on *GBC*, an *integer* kernel whose nullity and whose companion’s positivity are proved at every degree in [S]; P_5 is a minimality refinement sitting on top of a settled result.

4.2 Scaling to $d = 6$

Fitting kernels to the packets in range does not survive the move to $d = 6$. There are 29 packet constraints against 766 free monomials at the base fit depth, so a fitted kernel zeroes the packets it can see and breaks one truncation higher; the packet supply grows by about four constraints per degree, so no reachable truncation repairs it.

We therefore construct kernels instead of fitting them, from the antisymmetry of the residue integrand. Full invariance of P/D_d under one transposition yields 65 contour-exchangeable kernels, or 243 if every transposition is admitted; partial absorption raises the pool to 2297. That larger family is a shape rather than a theorem, so each candidate is put through the packet falsifier, which at depth 20 leaves 410. These are 410 distinct polynomials satisfying 17 linear relations, so they span a space of dimension 393; the redundancy is in the parameterisation, not in the truncation, and the series rank plateaus at the same 393 from depth 24 onwards.

4.3 The search at $d = 6$, and what would settle it

Since the chamber series is linear in the numerator, a positive gauge is a point of

$$\text{Feas}(T) = \{ t : \text{every coefficient of } Q_6 - \sum_i t_i K_i \text{ of degree } \leq T \text{ is } \geq 0 \}.$$

The kernel coefficients are a *fixed* finite set of variables, so raising T adds constraints and never variables: the $\text{Feas}(T)$ are nested decreasing. Two consequences matter. A feasible point at any finite T is only a finite verification. But $\text{Feas}(T) = \emptyset$ at a single T rules out a positive gauge in this span at *every* degree, because a constraint of total degree T involves only coefficients of degree $\leq T$, so truncating is an exact projection of the feasible set rather than an approximation of it.

The complementary condition is boundedness. If $\text{Feas}(T)$ is non-empty and bounded, the $\text{Feas}(T')$ for $T' \geq T$ are nested non-empty compacts, so their intersection is non-empty and a positive gauge exists. Boundedness is decided by the recession cone $C(T) = \{d : A_T d \leq 0\}$, which is trivial exactly when, by Stiemke’s lemma, there is $y > 0$ with $A_T^T y = 0$ — one linear programme.

Neither condition triggers up to depth 40. That rules nothing out in either direction: the feasible set is a non-empty unbounded polyhedron throughout.

4.4 The question is decidable at finite depth

What makes this worth continuing is that the sweep cannot run forever without answering.

Table 4: Both settlement conditions at $d = 6$, on the 393-dimensional kernel span.

T	constraints	Feas(T)	recession cone $C(T)$	time
24	67,952	non-empty	non-trivial	58 s
28	140,402	non-empty	non-trivial	150 s
32	266,538	non-empty	non-trivial	364 s
36	472,764	non-empty	non-trivial	676 s
40	793,228	non-empty	non-trivial	1542 s

Proposition 7. *Assume the K_i are residue-null and linearly independent. Then $C(T) = \{0\}$ for some finite T , and consequently exactly one of the following occurs at finite depth: either Feas(T) becomes empty, so no positive gauge exists in the span at any degree; or $C(T)$ becomes trivial while Feas(T) is non-empty, so a positive gauge exists and A_6 Chern positivity follows.*

Proof. First $C(\infty) = \{0\}$. Let $d \in C(\infty)$ and put $K = \sum_i d_i K_i$, so K is null and its chamber series is coefficientwise non-positive in every degree. Every chamber monomial β lies in exactly one packet, namely that of the multiset of $\alpha(\beta)$, and every packet sum of a null kernel vanishes; a sum of non-positive numbers equal to zero has every term zero, so the series vanishes identically. Multiplication by Φ is injective in the power series ring, so $K = 0$, and independence gives $d = 0$.

Now suppose $C(T) \neq \{0\}$ for all T and choose unit vectors $u_T \in C(T)$. The unit sphere is compact, so some subsequence converges to a unit vector u . For $T' \geq T$ we have $u_{T'} \in C(T') \subseteq C(T)$, and $C(T)$ is closed, so $u \in C(T)$ for every T . Then $u \in \bigcap_T C(T) = \{0\}$, contradicting $\|u\| = 1$. Hence $C(T^*) = \{0\}$ for some finite T^* , and the dichotomy follows from the nesting of Feas. $\square$

The hypothesis is where the care is needed. The argument uses vanishing packet sums, so it requires the K_i to be *genuinely* null, and the 410 were selected by the packet falsifier, which can refute nullity but never certify it — at $d = 6$ a filter run at depth 20 still passes kernels that move four packet sums at depth 22. Certifying the span by the degree-independent two-swap identity is therefore a prerequisite to either conclusion, not a refinement of it. Table 4 locates roughly where T^* lies; it does not establish either branch.

5 Assessment

What the computations suggest about proof strategy:

1. The τ -pairing reduction is *not* the right frame in general. Both its inequalities survive at $d = 6$ and both fail at $d = 7$ (§3), while Rimányi’s conjecture survives, so the machinery is specific to small d even though the statement is not. The prefix-positivity stepping stone fails one order earlier still.
2. Since §4 shows the numerator is free up to a residue-null kernel, the question worth asking is which representative needs the weakest certificate, or is nonnegative on its own. At $d = 5$ one is, and no certificate is needed at all. At $d = 6$ the question is open but no longer open-ended: it reduces to a dichotomy that must resolve at finite depth, and neither branch has triggered by depth 40.
3. The continuous relaxation is the right algorithmic vehicle. Integrality chokes on matrices past a few thousand rows, while the relaxed programme reaches 793,228 constraints in twenty-six minutes. Fractional coefficients cost nothing in rigour, because nullity is linear and the kernels are null by an algebraic identity, so any rational combination of them

is exactly null. What does need care is positivity, which must be recomputed in exact arithmetic rather than read off the solver at a tolerance.

4. Since the τ -orbit is not the right unit of cancellation, the open question is what is. Table 2 is suggestive: the negative mass is small, bounded and spread over few orderings, while the positive mass is large and concentrated in one. A bound on the negative part of $C(M)$ in terms of its largest term would settle the conjecture without needing any pairing at all, and nothing we have computed argues against one existing.

5.1 Open questions and sharp edges

5.1.1 Which positivity, in which basis?

Rimányi’s conjecture is a statement about the relative *Chern monomial* basis. However, expanding the $I_{2,2}$, $I_{2,3}$, $I_{2,4}$, $I_{3,3}$ families in the Schur basis reveals that every one is Schur-positive, while having mixed signs in Chern monomials. Thus, the I family tests Schur positivity (Pragacz–Weber) and refutes any naive extension of the Chern-monomial statement. Is Chern-monomial positivity special to Morin singularities, or the shadow of a statement about a basis interpolating between Chern monomials and Schur polynomials?

5.1.2 Why $\mathcal{Q}_8$ requires different mathematics

Between $d = 7$ and $d = 8$ the ambient dimension jumps from $34 \rightarrow 50$ and $\deg \mathcal{Q}_d$ goes $13 \rightarrow 22$. The saturation sequence prices out of computational reach. What $\mathcal{Q}_8$ would buy is the ability to analyze the individual components A_β , but Rimányi’s conjecture is already supported for A_8 and A_9 via restriction equations. This strongly argues for a different mathematical route to $\mathcal{Q}_d$ — such as localization on a resolution — rather than a faster Gröbner basis.

Reproducibility

Everything above is reproducible from a clean checkout of Chern++. Statements are separated throughout into finite exact verifications and proofs; the certificate machinery and the closed-form family arguments are the only components producing the latter. The annotated walkthrough is `src/examples.ipynb`, executed with its outputs.

A Computational Implementation

A.1 Scaling Saturation to $d = 7$

For a homogeneous ideal and degrevlex, a Gröbner basis saturates by v on dividing each element by the largest power of v dividing it. Both the placement of defect-zero coordinates and the sequence in which they are saturated changes computation time by orders of magnitude. The initial ideal is monomial, so the minimal primes are the minimal transversals of the generator supports, bypassing full primary decomposition and keeping the geometry exactly computable at $d = 7$.

A.2 The continuous relaxation, and the cost of exactness

At $d = 6$ a mixed-integer formulation stalls: branch-and-bound does not finish on the chamber matrix at usable depths. The relaxation does, and scales — the full boundary at depth 40 is 793,228 constraints on 393 variables and solves in about twenty-six minutes with HiGHS, against 58 seconds at depth 24, growing by roughly a factor 2.2 per four degrees.

The solver returns floating-point vertices. That is adequate for search and inadequate for a claim, so acceptance rationalises the solution and recomputes the gauged series exactly before reporting success. Recovery is deliberately partial: the constraint matrix is integral, so vertices are rational, but their denominators divide integer subdeterminants and can be large, and a float carries no record of which rational produced it. A candidate is accepted only if it reproduces the float to its own precision, and otherwise the run reports failure rather than substituting a nearby kernel for the one the solver found. Exact vertex extraction from the active constraint set would remove that gap and is not attempted here.

One measurement guides where exactness is cheap. Every quantity in the packet filter is an integer of magnitude at most about 10^7 , and float64 represents integers exactly below 2^{53} , so equality comparisons on the fast path are genuine rather than tolerances; a guard raises if any magnitude approaches that ceiling. Only fractional gauges force rational arithmetic, at a cost of some 25 to 65 times the float path.

B Numerical data

The counts below are rendered from the artifacts by `tools/render_tables.py` rather than transcribed, so they cannot drift from the pipeline.

Table 5: The mined chamber algebras. $\deg \mathcal{Q}_d = \dim \hat{N}_d - \binom{d}{2}$ is the codimension of the orbit closure; there are $d - 1$ chamber variables throughout.

	$\dim \hat{N}_d$	$\deg \mathcal{Q}_d$	$\mathcal{Q}_d$ terms	$\max \mathcal{Q}_d $	Vand. terms	N_d terms	$\max N_d $	N_d negative
A_1	0	0	1	1	1	1	1	0 (0%)
A_2	1	0	1	1	2	2	1	1 (50%)
A_3	3	0	1	1	6	6	1	3 (50%)
A_4	7	1	3	2	24	54	3	27 (50%)
A_5	13	3	19	8	120	814	18	414 (51%)
A_6	22	7	418	233	720	26618	800	13309 (50%)
A_7	34	13	14586	37880	5040	1210404	196803	605234 (50%)

Table 6: The orbit closure $\mathcal{O}_d \subset \hat{N}_d$. “mult” is the range of component multiplicities, “deg Hilb” the degree of the Hilbert numerator. For $d \leq 3$ the orbit fills $\hat{N}_d$ and there is no degeneration.

	$\dim \hat{N}_d$	$\dim \mathcal{O}_d$	codim	$\deg \mathcal{O}_d$	components	mult	deg Hilb	$\mathcal{Q}_d$
A_1	0	0	0	1	—	—	0	1
A_2	1	1	0	1	—	—	0	1
A_3	3	3	0	1	—	—	0	1
A_4	7	6	1	2	2	1–1	2	irreducible
A_5	13	10	3	6	5	1–2	5	1 · 2
A_6	22	15	7	55	44	1–2	12	irreducible
A_7	34	21	13	957	572	1–8	21	irreducible

References

- [1] Bérczi, L., & Szenes, A. (2012). Thom polynomials of Morin singularities. *Annals of Mathematics*, 175(2), 567–629.
- [2] Rimányi, R. (2001). Thom polynomials, symmetries and incidences of singularities. *Inventiones mathematicae*, 143(3), 499–521.

Table 7: Negative Laurent coefficients against the positive mass they sit in, on the truncation shown, and the Chern coefficients at $\ell = 0$ they cancel into.

	$\deg \leq$	A_β	negative	%	least	neg. mass	of pos.	$\min C(M)$
A_4	12	305	0	0.0%	—	0	0.00%	1
A_5	12	687	7	1.0%	−1	7	0.03%	1
A_6	10	526	6	1.1%	−1	6	0.09%	1
A_7	10	657	13	2.0%	−2	14	0.19%	1

- [3] Pragacz, P., & Weber, A. (2007). Positivity of Schur function expansions of Thom polynomials. *Fundamenta Mathematicae*, 195, 85–100.